

L-Aspartate *de novo* and Selected Salvage Routes to NAD⁺

A scope-limited review of the routes that converge on nicotinate mononucleotide (NaMN) and complete NAD⁺ through deamido-NAD, plus the topologically distinct NAMPT→NMN direct-completion route.

1. Executive summary

Nicotinamide adenine dinucleotide (NAD⁺) is assembled from a small pyridine mononucleotide (NaMN or NMN) by a two-reaction "completion" chemistry — adenylation of the mononucleotide to a dinucleotide, then installation of the amide nitrogen. The system reviewed here is the set of routes that (i) build or recover the **deamidated** mononucleotide **nicotinate mononucleotide (NaMN)**, and (ii) carry NaMN through **deamido-NAD (nicotinate adenine dinucleotide, NaAD)** to NAD⁺. NaMN is made by four routings of three chemistries: *de novo* from **L-aspartate** (NadB → NadA → NadC); by **Preiss–Handler** phosphoribosylation of nicotinate (PncB) either directly or after **nicotinamide deamidation** by PncA; or by **deamidation of NMN** (PncC). All three converge on a single obligatory node: NaMN must be adenylylated (NadD, or a bifunctional NMNAT) to deamido-NAD and then amidated by **NAD synthetase (NadE/NADSYN)** to NAD⁺.

Standing apart from this deamidated arm is one **amidated** route that is included here because it shares the same terminal adenylyltransferase family but uses a different completion topology: **NAMPT** phosphoribosylates nicotinamide to **NMN**, and NMN is adenylylated **directly** to NAD⁺ by NMNAT, with **no NAD synthetase step**. The central organizing principle of the entire system is therefore *when the amide nitrogen is installed*: **late** (deamidated arm, NadE-dependent) versus **early** (NAMPT arm, NadE-independent).

The review deliberately excludes the L-tryptophan/kynurenine *de novo* route (owned by a separate module), nicotinate degradation, transhydrogenases, NAD kinase/NADP formation, and NAD-consuming reactions.

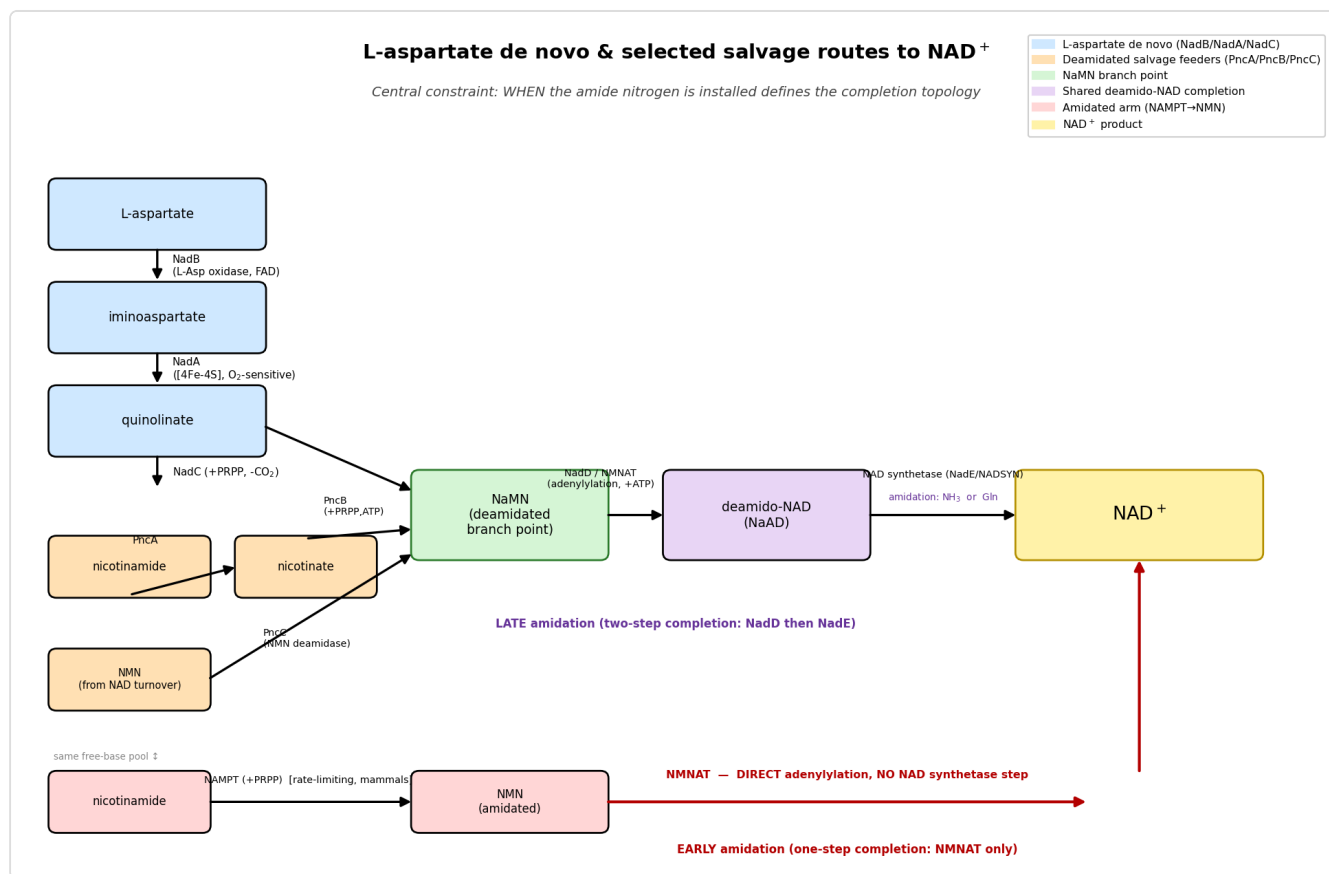


Figure 1. Architecture of the two NAD⁺ completion topologies

Figure 1. The scope-limited route set. Four sources feed the deamidated branch point **NaMN** (L-aspartate *de novo* via NadB/NadA/NadC; nicotinate via PncB; nicotinamide via PncA→PncB; NMN via PncC), which is obligatorily completed in **two steps** — adenylylation to deamido-NAD (NadD or NMNAT) then amidation by NAD synthetase (NadE/NADSYN). The **amidated arm** (bottom, red) is topologically separate: NAMPT makes NMN already bearing the amide, and NMNAT adenylylates it **directly** to NAD⁺ with no NAD synthetase step. The organizing variable is the *timing of amide-nitrogen installation* — late (deamidated arm) versus early (amidated arm).

2. Definition and biological boundaries

2.1 What is inside the boundary

The system comprises exactly these reactions:

A. NaMN production

1. **L-aspartate *de novo***: L-aspartate → iminoaspartate (**NadB**, L-aspartate oxidase) → quinolinate (**NadA**, quinolinate synthase) → NaMN (**NadC**, quinolinate phosphoribosyltransferase, decarboxylating).
2. **Direct Preiss–Handler entry**: nicotinate → NaMN (**PncB**, nicotinate phosphoribosyltransferase).
3. **Nicotinamide deamidation then Preiss–Handler**: nicotinamide → nicotinate (**PncA**, nicotinamidase) → NaMN (**PncB**).
4. **NMN deamidation**: NMN → NaMN (**PncC**, NMN deamidase).

B. Deamidated completion (shared)

5. NaMN → deamido-NAD (**NadD** bacterial family, or bifunctional **NMNAT**).
6. deamido-NAD → NAD⁺ (**NadE/NADSYN**, ammonia- or glutamine-dependent).

C. Amidated direct completion (separate topology)

7. nicotinamide → NMN (**NAMPT**).
8. NMN → NAD⁺ directly (**NMNAT**).

2.2 What is outside the boundary (and often confused with it)

- **L-tryptophan / kynurenine *de novo* route**. In fungi and animals, quinolinate — the same intermediate produced by **NadA** — is generated instead from tryptophan via kynurenine (TDO/IDO, KMO, kynureninase **KYNU**, **HAAO**). This produces the identical NaMN branch point but by entirely different upstream chemistry and is treated in the separate `kynurenine_nad_de_novo` module. It is important not to double-count quinolinate → NaMN → NAD⁺: **NadC** and the completion enzymes are shared, but the aspartate and tryptophan arms are parallel, mutually alternative sources of quinolinate. Human Congenital NAD Deficiency Disorder maps to the *tryptophan* arm (**KYNU**, **HAAO**) **and** to the shared completion enzyme **NADSYN1**

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- **Nicotinate degradation** (the *nic* catabolic cluster) — consumes, rather than salvages, nicotinate.
- **Pyridine-nucleotide transhydrogenases, NAD kinase** (NAD⁺ → NADP⁺), and **NAD-consuming enzymes** (sirtuins, PARPs, ADP-ribosyl cyclases, bacterial DNA ligase). These define the demand side and the NADP pool, not NAD⁺ synthesis, and are excluded.
- **Nicotinamide riboside (NR) salvage** via nicotinamide riboside kinases (**NRK1/2**) feeds NMN and is adjacent, but is not itemized in this scope's provisional outline; it is mentioned only where it clarifies the NMN node.

2.3 Competing definitions in the literature

Two framing conventions coexist. The **microbial** literature (Osterman, Rodionov, Raffaelli and colleagues) treats NAD homeostasis as a **pyridine nucleotide cycle**: NAD is consumed to nicotinamide/NMN and *recycled* through PncA/PncB/PncC back to NaMN

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The **mammalian** literature emphasizes the **NAMPT-centred salvage** loop (nicotinamide → NMN → NAD⁺) as the dominant flux

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These are not contradictory: they describe organisms that install the amide nitrogen at opposite ends of the pathway. A recurring source of confusion is the word "salvage," which is used for both the deamidated (PncA/PncB/PncC) and amidated (NAMPT/NRK) routes even though they complete NAD⁺ by different topologies.

3. Mechanistic overview

3.1 The two completion topologies

The unifying insight is that NAD⁺ completion is a **mononucleotide → dinucleotide adenylation** followed (or not) by an **amidation**:

- **Deamidated arm.** NaMN (carries a carboxylate at the pyridine 3-position) is adenylylated to deamido-NAD (NaAD), which still carries the carboxylate. A dedicated **NAD synthetase (NadE)** then converts the carboxylate to the amide using ATP activation and an ammonia donor. Two chemical steps of nitrogen handling are required overall (the pyridine ring enters already deamidated).
- **Amidated arm.** NMN already carries the amide. A single adenylation by **NMNAT** yields NAD⁺ directly. No NadE step exists or is needed.

Consequently, **the oxidation/deamidation state of the mononucleotide dictates the number of downstream steps**: anything at the NaMN state is committed to the two-step NadD + NadE completion; anything at the NMN state can be completed in one step. This is the single most important constraint in the system (Section 6).

3.2 Obligatory, conditional, and accessory steps

- **Obligatory for any organism using this system:** the adenylation step (NadD or NMNAT) and, for the deamidated arm, the amidation step (NadE). $\text{NaMN} \rightarrow \text{deamido-NAD} \rightarrow \text{NAD}^+$ has no known bypass.
- **Conditional (route-selecting):** which NaMN source operates depends on genotype and nutrient supply. *De novo* NadABC is used when no exogenous pyridine is available; Preiss–Handler (PncB) is used when nicotinate is available; PncA precedes PncB when the supplied vitamer is nicotinamide; PncC operates where NMN accumulates from NAD turnover.
- **Accessory / recycling:** PncA, PncB, and PncC are dispensable in a minimal-medium *de novo* organism but essential for pyridine-nucleotide cycling and for growth on specific vitamers. Regulatory proteins (NiaR, NrtR, Rex) tune expression but are not catalytic.

3.3 Step-by-step

1. **L-aspartate oxidation (NadB).** An FAD-dependent desaturating dehydrogenase converts L-aspartate to iminoaspartate, reducing FAD; the flavin is reoxidized by **fumarate anaerobically** and by **molecular O₂ aerobically**, the latter releasing H₂O₂ as an adventitious by-product (eliminating NadB turnover cut endogenous H₂O₂ by ~25–30 % in an E. coli scavenger-null strain)

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NadB is homologous to the flavoprotein subunit of succinate dehydrogenase/fumarate reductase — a deeply conserved respiratory-flavoprotein lineage. This redox flexibility lets the *de novo* arm's first step operate across oxygen tensions, complementing the O₂-**sensitivity** of the downstream [4Fe-4S] NadA.

2. **Quinolinate synthase (NadA).** Condenses iminoaspartate with dihydroxyacetone phosphate (DHAP) and cyclizes/dehydrates to quinolinate

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NadA carries a **[4Fe-4S]²⁺ cluster** ligated by only three conserved cysteines, leaving a unique non-cysteinyll fourth iron that binds substrate/intermediate and is central to catalysis

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The cluster makes NadA **oxygen-sensitive**, a physiologically important liability.

3. **Quinolate phosphoribosyltransferase (NadC).** Transfers the phosphoribosyl group from PRPP to quinolate and decarboxylates, giving NaMN + CO₂. This is a type-II PRTase and is the point at which the aspartate and tryptophan *de novo* arms merge.
 4. **Preiss-Handler / deamidation feeders.** PncA (nicotinamidase) hydrolyzes nicotinamide → nicotinate + NH₃; PncB (nicotinate PRTase) uses PRPP and ATP to make NaMN (ATP hydrolysis raises the effective equilibrium toward NaMN); PncC deamidates NMN → NaMN directly.
 5. **NaMN adenylation (NadD or NMNAT).** ATP + NaMN → NaAD + PPi.
 6. **Amidation (NadE/NADSYN).** NaAD is adenylylated on its carboxylate (forming an NaAD-adenylate), then attacked by ammonia (ammonia-dependent NadE) or by ammonia liberated in-situ from glutamine (glutamine-dependent NADSYN), giving NAD⁺ + AMP + PPi.
 7. **NAMPT arm.** NAMPT + PRPP convert nicotinamide → NMN; NMNAT adenylylates NMN → NAD⁺ directly.
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4. Major molecular players and active assemblies

Step	Enzyme (gene)	Family / fold	Cofactor / donor	Notable assembly features
Asp oxidation	L-aspartate oxidase (nadB)	SDH/FRD flavoprotein	FAD; fumarate (anaerobic) or O ₂ (aerobic, H ₂ O ₂)	Interacts with NadA (not species-specific); O ₂ use adventitious (P 18959769) (P 20149100)
Quinolate synthesis	Quinolate synthase (nadA)	[4Fe-4S] enzyme	[4Fe-4S] ²⁺ , 3-Cys + unique Fe	O ₂ -sensitive; three-domain pseudo-symmetric fold (P 18959769) (P 31390192)
Quinolate → NaMN	Quinolate PRTase (nadC)	Type-II PRTase	PRPP; Mg ²⁺	Often hexameric; decarboxylating
Nicotinamide → nicotinate	Nicotinamidase (pncA)	Cys-His-Asp metalloamidohydrolase	Zn/Mn/Fe	Same enzyme = pyrazinamidase in mycobacteria (P 8726014) (P 18201201)
Nicotinate → NaMN	Nicotinate PRTase (pncB)	Type-II PRTase, ATPase-assisted	PRPP; ATP	Facultative ATPase (Ap4 synthesis/degradation) (P 28416276)
NMN → NaMN	NMN deamidase (pncC)	Novel Ser/Lys amidohydrolase (CinA C-term)	none	Distinct fold; often fused to COG1058 ADP-ribose pyrophosphatase

Step	Enzyme (gene)	Family / fold	Cofactor / donor	Notable assembly features
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NaMN → NaAD	NaMN adenylyltransferase (nadD)	Nucleotidyltransferase α/β phosphodiesterase (HxGH)	ATP; Mg ²⁺	Bacterial NadD is nicotinate-selective; dimeric in some species (P 11704676) (P 16784754)
NaMN/NMN → (Na)AD	NMNAT (NMNAT1/2/3)	Same NT superfamily	ATP; Mg ²⁺	Dual-specificity; 3 compartmentalized human isoforms (P 16118205) (P 17402747)
NaAD → NAD ⁺	NAD synthetase (nadE / NADSYN1/2)	ATP-pyrophosphatase (N-type) ± CN-hydrolase	ATP; NH ₃ or Gln	Glutamine form adds glutaminase domain + ~40 Å ammonia tunnel (P 12547821) (P 22280445)
Nicotinamide → NMN	NAMPT (NAMPT / PBEF/visfatin)	Type-II PRTase (dimeric)	PRPP; ATP-assisted	Rate-limiting in mammals; FK866-inhibited (P 35903330) (P 28416276)

Active-assembly highlights.

- **NadA/NadB coupling.** The two *de novo* enzymes form a functional pair; cross-species reconstitution (B. subtilis NadA with E. coli NadB) works, indicating a conserved interaction interface for iminoaspartate channelling

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Iminoaspartate is unstable, so hand-off limits its escape.

- **NadD as a fold archetype.** NadD, NMNAT, and phosphopantetheine adenylyltransferase share the nucleotidyltransferase α/β phosphodiesterase superfamily with the (T/H)xGH motif; bacterial NadD and human NMNAT are structurally similar yet pharmacologically separable, which is why NadD is an antibacterial target

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- **Glutamine-dependent NadE as a channeling machine.** The synthetase and glutaminase active sites are separated by a ~40 Å ammonia tunnel with three constriction points that gate transfer and enforce kinetic synergism between the two active sites

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5. Evolutionary and cell-biological variation

5.1 Across major lineages

- **Bacteria (and plants, and many archaea).** Predominantly use the **L-aspartate *de novo*** arm (NadB/NadA/NadC) plus NadD (nicotinate-specific) and NadE. The amide nitrogen is installed **late**. Deamidated salvage (PncA/PncB/PncC) recycles pyridines back to NaMN. Comparative-genomic reconstructions across hundreds of bacteria show extensive lineage-specific presence/absence of these genes and multiple transcriptional strategies (NiaR in Firmicutes/Thermotogales; NrtR/Nudix regulators broadly; Rex for redox balance)

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- **Fungi and animals.** Lack NadB/NadA; make quinolinate from **tryptophan** (outside this scope) and rely heavily on **amidated NAMPT salvage**, with NMN completed directly by **NMNAT** — the amide nitrogen is installed **early**. Humans have both an ammonia-dependent (NADSYN2-like) and glutamine-dependent (NADSYN1) NAD synthetase for the deamido-NAD produced by their tryptophan/ Preiss–Handler flux

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- **Pathogen reductive loss.** *Shigella*/EIEC illustrate pathoadaptive loss of the *de novo* arm: silencing of *nadA/nadB* yields a nicotinic-acid–requiring (Nic[−]) phenotype, apparently because quinolinate attenuates invasion

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This is a clean natural experiment showing the *de novo* arm is dispensable when salvage substrate is available.

- **Mobile/large genetic elements.** Some gut "jumbo" phages encode complete NAD synthesis gene sets, an extreme case of horizontal acquisition of this module

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5.2 Enzyme-family variation that achieves the same outcome

- **Adenylyltransferase families:** bacterial **NadD** (nicotinate-selective, deamidated arm only) versus eukaryotic **NMNAT** (dual NaMN/NMN specificity, serves both arms). These are the "alternative adenylyltransferase families" of the brief.
- **NAD synthetase nitrogen donors: ammonia-dependent** (synthetase core only) versus **glutamine-dependent** (added CN-hydrolase glutaminase domain + ammonia tunnel). The glutamine dependence is a modular add-on: mutating the glutaminase catalytic Cys reverts NADSYN1 to ammonia-only behaviour

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5.3 Cell-type, compartment, and physiological state (eukaryotes)

The three human NMNAT isoforms are compartmentalized — NMNAT1 (**nucleus**), NMNAT2 (**Golgi/cytosol**), NMNAT3 (**mitochondria**) — with distinct kinetics, metal preferences, and substrate-binding order, arguing for **compartment-specific NAD pools** rather than one freely mixing cellular pool

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NMNAT2 is a labile axon-survival factor whose loss triggers Wallerian-type axon degeneration (a major reason this branch is biomedically prominent). NAMPT is rate-limiting and stress-regulated, coupling NAD⁺ supply to metabolic and inflammatory state

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5.4 Deepest plausible origin and what is ancient vs. derived

NAD⁺ itself is among the very small set of coenzymes assigned to the core metabolism of the **last universal common ancestor (LUCA)** — alongside ADP and pyridoxal phosphate — and all three have been shown to be catalytically active in their metabolic reactions **in the absence of enzymes**, implying a prebiotic contribution to abiogenesis

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The pressure to synthesize and regenerate NAD is therefore essentially as old as cellular life, and the **completion chemistry** — mononucleotide → dinucleotide adenylylation followed by amidation — is best read as the ancient, conserved core of the system.

Reading the parts by apparent age:

- **Ancient / conserved:** (i) the adenylylation step, catalyzed by the nucleotidyltransferase α/β -phosphodiesterase superfamily that also produced phosphopantetheine adenylyltransferase — a fold shared by NadD and NMNAT and therefore predating the bacterial/eukaryotic split; (ii) NAD synthetase, an ancestral member of the ATP-pyrophosphatase (N-type amidotransferase) clan; (iii) the **L-aspartate *de novo* arm** — NadB belongs to the deeply conserved succinate-dehydrogenase/fumarate-reductase flavoprotein family and works anaerobically (fumarate) as readily as aerobically

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while NadA's **[4Fe-4S]** chemistry is a hallmark of ancient, anaerobe-compatible metallo-enzymes. The aspartate route's anaerobic compatibility fits an early, low-oxygen biosphere.

- **Later elaborations / replacements / losses:** (i) the **glutaminase domain + ammonia tunnel** appended to NAD synthetase is a modular add-on to an ammonia-dependent core (mutating the glutaminase Cys reverts human NADSYN1 to ammonia-only;

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and phylogenetics identify a *poorly* synergistic glutamine-dependent enzyme (*T. maritima*) as closer to the ancestral state, with tight coupling being derived

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(ii) the **amidated NAMPT→NMN→NMNAT** topology, dominant in animals, is a lineage where the *de novo* aspartate arm was lost and replaced by tryptophan-derived quinolinate plus heavy amidated salvage; (iii) **reductive losses** of *de novo* genes (*Shigella nadA/nadB*; obligate host-associated bacteria) recur wherever pyridine vitamers are reliably available.

Best representatives for the ancestral role. Because these families expanded, the least specialized members are the most informative: for the adenylyltransferase fold, **bacterial NadD** (single-domain, nicotinate-specific) better reflects the ancestral adenylylation than the compartment-specialized, dual-specificity eukaryotic NMNAT isoforms; for the amidation step, a **poorly-coupled glutamine-dependent NadE** such as *T. maritima*'s, or a plain **ammonia-dependent NadE**, better represents the ancestral synthetase than the highly synergistic human/mycobacterial enzymes

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6. Constraints, dependencies, and failure modes

6.1 Ordering constraints (hard)

1. **Amide-timing determines topology.** A nucleotide at the NaMN state can *only* reach NAD⁺ via NadD → NadE (two nitrogen-handling operations). A nucleotide at the NMN state reaches NAD⁺ via NMNAT alone. There is no productive "NaMN → NAD⁺ direct amidated adenylylation" and no "NMN → deamido-NAD → NAD⁺" *within the completion step* — the

only way to move NMN into the deamidated arm is to **deamidate it first (PncC)**. This mutual exclusivity is the backbone constraint of the system.

2. **Adenylylation precedes amidation.** NadE acts on deamido-NAD, not on NaMN; the ribose-adenylyl dinucleotide must exist before the ring amide can be installed. Order is fixed.
3. **NadB precedes NadA precedes NadC.** Iminoaspartate (NadB product) is the NadA substrate; quinolinate (NadA product) is the NadC substrate. The instability of iminoaspartate enforces tight kinetic/physical coupling of NadB–NadA.
4. **PncA precedes PncB** when the entering vitamer is nicotinamide (must be deamidated to nicotinate before phosphoribosylation by PncB).

6.2 Substrate/compartment specificity

- Bacterial **NadD is nicotinate-selective**: it will not efficiently complete amidated NMN, so bacteria relying on NadD must deamidate (PncC) any NMN they wish to reuse

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Eukaryotic **NMNAT is permissive** and therefore removes this constraint, which is part of why the amidated arm dominates in animals.

- **O₂ sensitivity of NadA** restricts robust *de novo* flux under aerobiosis unless the cluster is protected/repared; this is one plausible selective pressure favouring salvage in some aerobes.

6.3 Failure modes

- **Loss of completion enzymes is lethal/near-lethal.** NadD and NadE are essential in most bacteria (validated antibacterial targets;

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In humans, biallelic **NADSYN1** loss-of-function causes **Congenital NAD Deficiency Disorder** (VACTERL-like vertebral, cardiac, renal, limb malformations; NAD pools partially rescuable by nicotinamide supplementation), demonstrating that the amidation node is developmentally indispensable

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- **Loss of a NaMN source is conditionally lethal.** *nadA/nadB* silencing is tolerated only when nicotinate/nicotinamide is supplied (Nic⁻ phenotype;

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- **NMN toxicity/regulatory coupling.** Accumulated NMN can inhibit essential enzymes (e.g., DNA ligase in *Salmonella*); PncC recycling relieves this, linking the deamidated arm to genome maintenance

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6.4 Evidence that rules out otherwise-plausible paths

- There is no enzyme that amidates NaMN directly to NMN in this system; the amide is installed either at the free base (NAMPT uses nicotinamide) or at the dinucleotide (NadE on NaAD) — never on the deamidated mononucleotide. The strict NaMN→NaAD→NAD⁺ order is supported by the substrate specificity of NadE (acts on NaAD) and of NadD (acts on NaMN)

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7. Controversies and open questions

1. **NadA catalytic mechanism and cluster ligation.** The role of the unique non-cysteinyll iron and the identity of on-pathway intermediates remain partly unsettled; a crystallographically trapped species has been argued to be an intermediate but its assignment is debated

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How the O₂-sensitive cluster is protected/repared in aerobes is unresolved.

2. **Regulation of the glutaminase↔synthetase coupling in NadE.** Kinetic synergism and ammonia-tunnel gating are documented, but the degree of "leak" (uncoupled glutamine hydrolysis) varies by organism, and an *ancestral*, poorly-synergistic glutamine-dependent enzyme (*T. maritima*) has been proposed — implying the tight coupling seen in *M. tuberculosis*/human is a derived optimization

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Whether the ancestral NadE was ammonia- or glutamine-dependent is not fully settled.

3. **One NAD pool or many (eukaryotes).** Compartmentalized NMNAT isoforms argue for distinct nuclear/Golgi/mitochondrial pools, but the extent of inter-compartment NAD/NMN transport is still being mapped

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4. **Moonlighting activities.** NAMPT and PncB/NAPT catalyze a facultative ATPase / adenosine-5'-tetraphosphate (Ap₄) metabolism of uncertain physiological weight

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 the PncC-associated COG1058 domain is an ADP-ribose pyrophosphatase whose functional link to NAD recycling is proposed but not fully defined
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5. **Organism-mixing caution.** Much mechanism is stitched from different organisms (*B. subtilis* NadA, *S. aureus*/*B. anthracis* NadD, *M. tuberculosis* NadE, human NMNAT/NADSYN, *S. typhimurium* PncC). Substrate specificity and essentiality are not safely transferable across these without checking — a recurring interpretive risk in this literature.

8. Key references

- Marinoni et al. 2008, *FEBS J* — *B. subtilis* NadA/NadB; NadA [4Fe-4S], 3-Cys ligation; cross-species NadA–NadB interaction.

P 18959769

- Korshunov & Imlay 2010, *Mol Microbiol* — NadB uses fumarate (anaerobic) / O₂ (aerobic, H₂O₂ by-product); homology to fumarate reductase flavoprotein.

P 20149100

- Dherbassy, Mayer & Moran 2024 — NAD among LUCA-essential coenzymes active non-enzymatically; prebiotic origin of NAD chemistry.

P 39292780

- Esakova et al. 2019, *Biochemistry* — NadA condenses DHAP + iminoaspartate; trapped putative intermediate.

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- Di Martino et al. 2013 — *Shigella*/EIEC pathoadaptive silencing of *nadA/nadB*, Nic[−] phenotype.

P 24120364

- Olland et al. 2002, *JBC* — *B. subtilis* NadD structure; nicotinate specificity; dimeric functional unit.

P 11704676

- Han et al. 2006 — *S. aureus* NaMNAT–NaAD complex; conserved NT fold; adenylation of NaMN → NaAD.

P 16784754

- Sorci et al. 2009 / Huang et al. 2010 — NadD as essential, druggable antibacterial target; inhibitor complexes.

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- Hara et al. 2003, *JBC* — human NADSYN1 (Gln+NH₃) vs NADSYN2 (NH₃-only); CN-hydrolase domain confers glutamine dependence (Cys-175).

[P 12547821](#)

- Chuenchor et al. 2012 / 2020 — *M. tuberculosis* glutamine-dependent NadE; ~40 Å ammonia tunnel and its regulation.

[P 22280445](#)

[P 31911602](#)

- Resto et al. 2009 — ancestral, poorly-synergistic glutamine-dependent NAD synthetase (*T. maritima*).

[P 19647806](#)

- Teixeira et al. 2021 — NadE among validated multifunctional anti-TB targets.

[P 33372865](#)

- Galeazzi et al. 2011 — identification of *pncC*/NMN deamidase; broadly conserved amidohydrolase family.

[P 21953451](#)

- Sorci et al. 2014 — PncC is a Ser/Lys serine-amidohydrolase (novel fold).

[P 24530526](#)

- Cialabrini et al. 2013 — PncC-fused COG1058 is an ADP-ribose pyrophosphatase; recycling link.

[P 23776507](#)

- Cheng & Roth 1995 — pyridine nucleotide cycle; PncC is the predominant NMN→NaMN recycling reaction; NMN toxicity to DNA ligase.

[P 7592458](#)

- Frothingham et al. 1996 / Zhang et al. 2008 — *E. coli* and *M. tuberculosis* PncA (nicotinamidase = pyrazinamidase).

[P 8726014](#)

[P 18201201](#)

- Berger et al. 2005 / Sorci et al. 2007 — three compartmentalized human NMNAT isoforms; kinetics and dual NaMN/NMN specificity.

[P 16118205](#)

[P 17402747](#)

- Zhu et al. 2022 / Wen et al. 2024 — NAMPT as rate-limiting mammalian salvage enzyme; drug target.

[P 35903330](#)

[P 38244254](#)

- Amici et al. 2017 — NAMPT/NAPT facultative ATPase and Ap4 metabolism.

[P 28416276](#)

- Rodionov et al. 2008 (×2) / Ravcheev et al. 2012 — NiaR, NrtR, and Rex regulation of NAD metabolism across bacteria.

[P 18276644](#)

[P 18276643](#)

[P 22210771](#)

- Szot et al. 2021 and related — Congenital NAD Deficiency Disorder (KYNU/HAAO/NADSYN1).

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[P 35491967](#)

[P 36649848](#)

[P 42153968](#)

- Li et al. 2024 — jumbo phages encoding complete NAD synthesis gene sets.

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Limitations of this review

Mechanistic detail is unevenly distributed across model organisms, and several claims are inferred from comparative genomics rather than direct biochemistry. Structural snapshots of NadA intermediates and of the NadE ammonia tunnel are powerful but come from single organisms. The amidated (NAMPT/NMNAT) arm is included for topological contrast but its upstream tryptophan-derived quinolinate flux is deliberately not treated here. Quantitative flux partitioning among the four NaMN sources in any single organism under defined conditions remains an open experimental target.

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